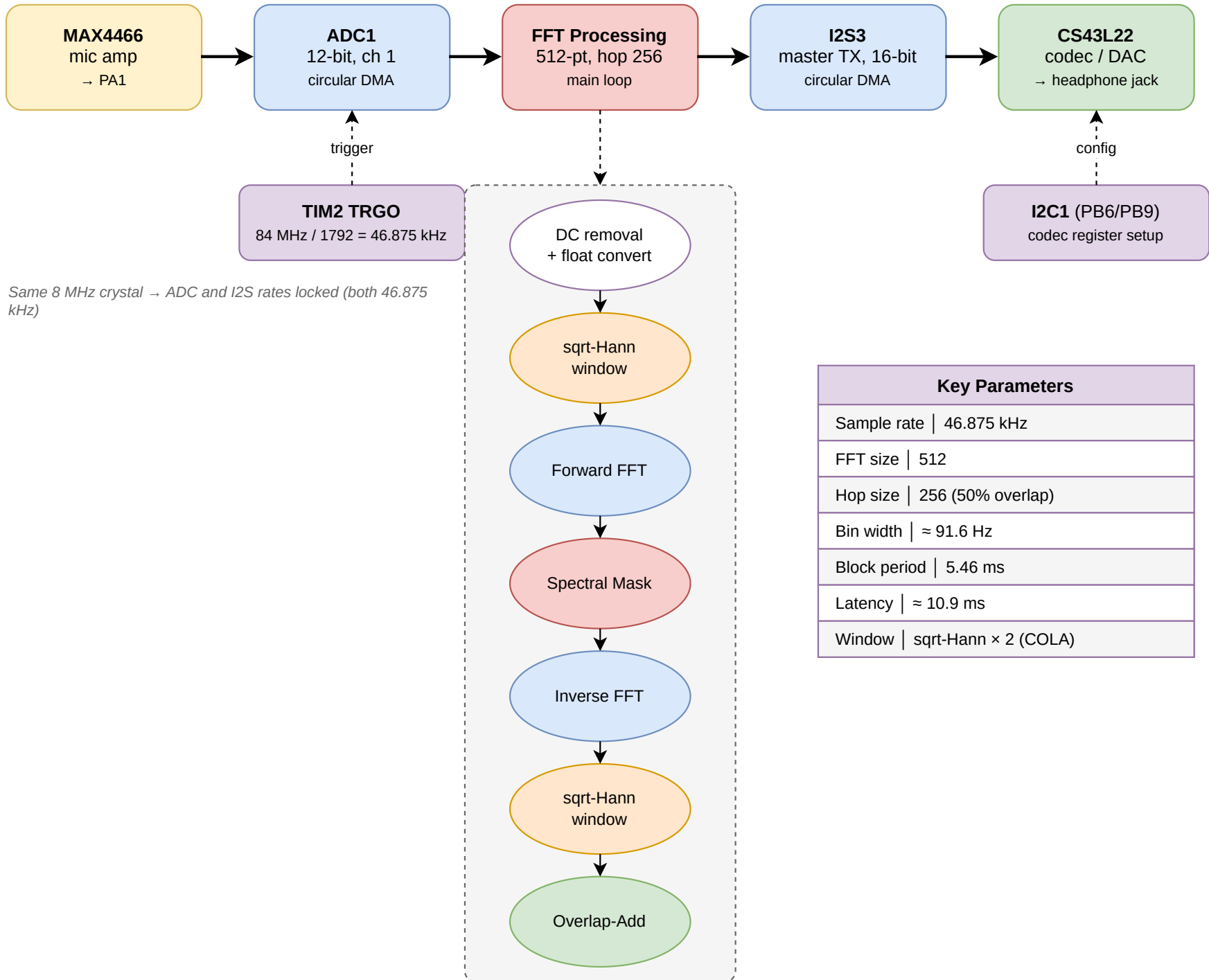
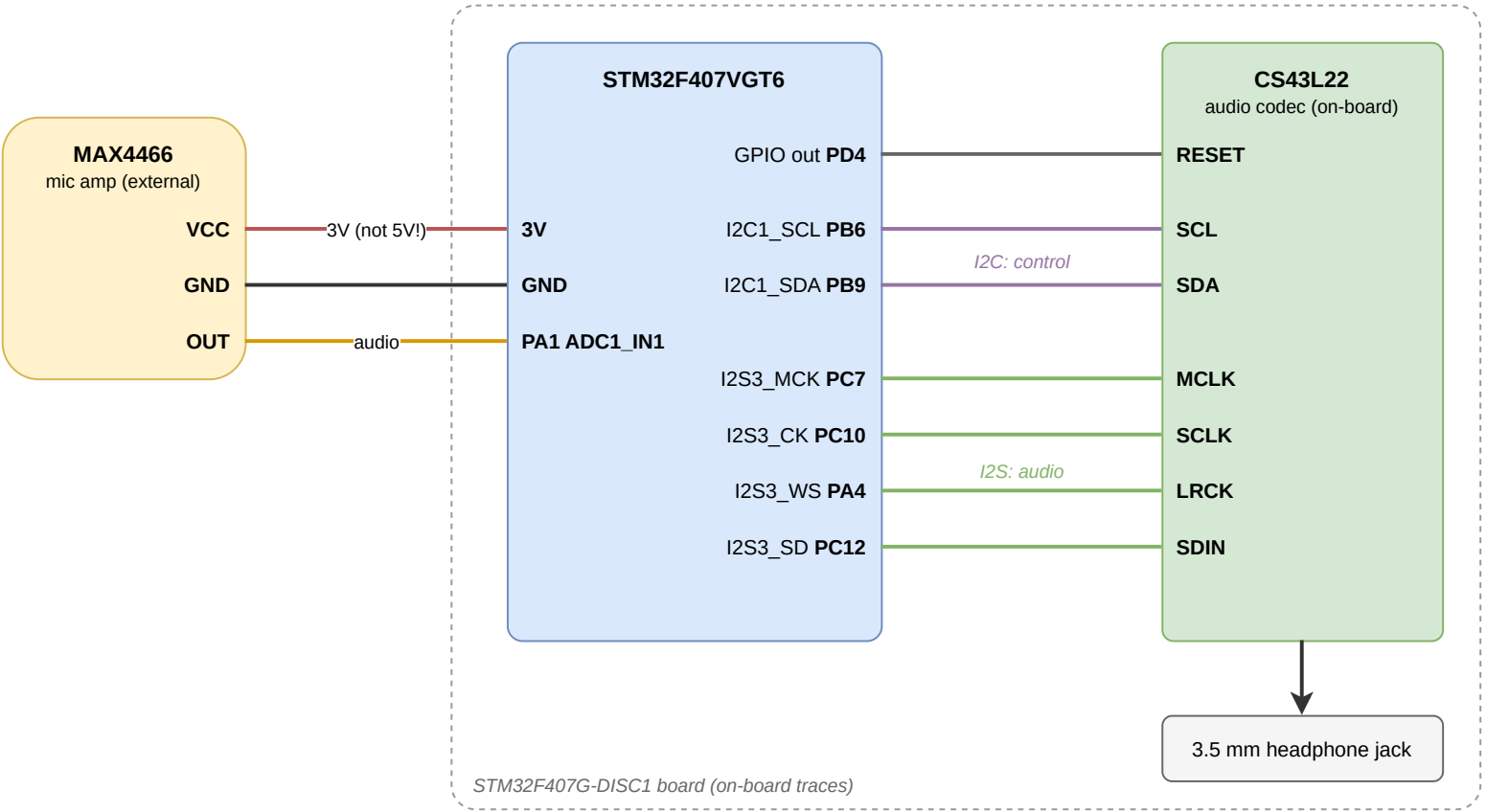


System Overview



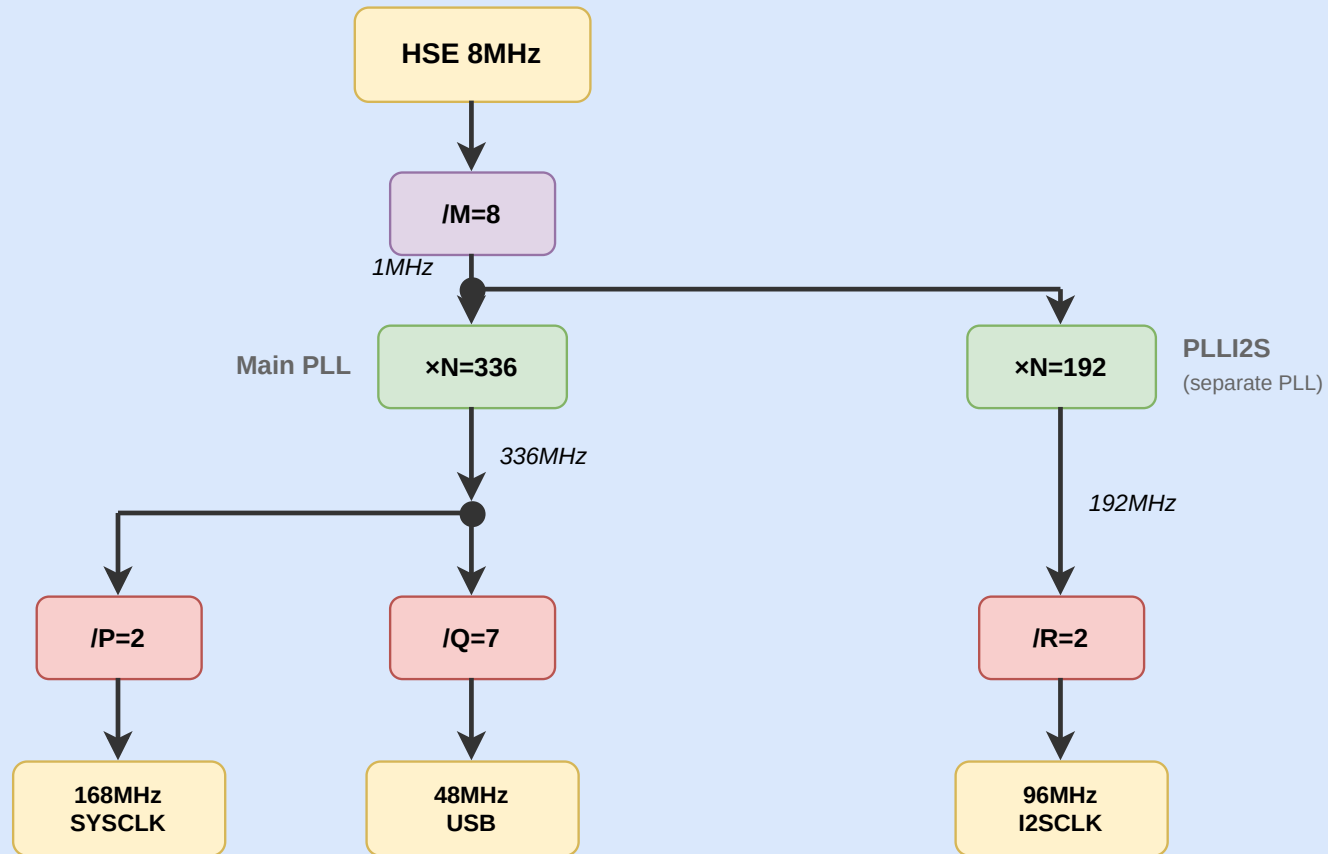
Pin	Function	Peripheral
PA1	ADC Input	ADC1_IN1
PD4	Codec Reset	GPIO Output
PB6	I2C Clock	I2C1_SCL
PB9	I2C Data	I2C1_SDA
PA4	I2S WS	I2S3_WS
PC7	I2S MCLK	I2S3_MCK
PC10	I2S Clock	I2S3_CK
PC12	I2S Data	I2S3_SD

Wiring



Only the mic is wired by hand. Codec, I2C, I2S and reset are PCB traces on the Discovery board.
Power the MAX4466 from **3V**: its output idles at VCC/2 and swings to VCC, and PA1 must stay below 3.6 V.
Keep the three jumper wires short and twisted to reduce hum pickup.

System Clock Tree



Bus Clocks	
SYSCLK	168 MHz
AHB (HCLK)	168 MHz (÷1)
APB1	42 MHz (÷4)
APB1 Timers	84 MHz (×2) ← TIM2
APB2	84 MHz (÷2)
ADC clock	21 MHz (APB2 ÷4) ← ADC1
APB2 Timers	168 MHz (×2)

I2S3 prescaler ÷8
16-bit, MCLK on: $F_s = 96\text{MHz} / (256 \times 8)$

$F_s = 46.875\text{ kHz}$
closest to 48k, CubeMX error -2.34%

TIM2 TRGO = $84\text{MHz} / (1791+1)$
= 46.875 kHz exactly → ADC1 trigger
same HSE crystal → locked to I2S

Peripheral Settings

ADC1 Configuration

Resolution	12-bit
Channel	ADC1_IN1 (PA1)
Trigger Source	TIM2 TRGO
Sampling Time	15 cycles
DMA Mode	Circular
DMA Buffer	512 × uint16 (256 samples × 2 halves)

TIM2 Configuration (ADC Trigger)

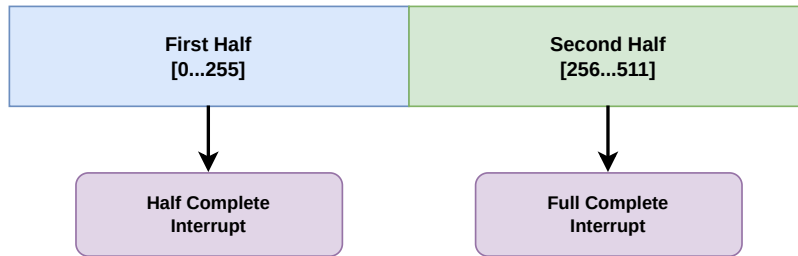
Clock Source	APB1 Timer (84 MHz)
Prescaler	0
Period	1791
Output Frequency	84MHz / 1792 = 46.875 kHz (matches I2S)
Trigger Output	Update Event (TRGO)

I2S3 Configuration (Audio Output)

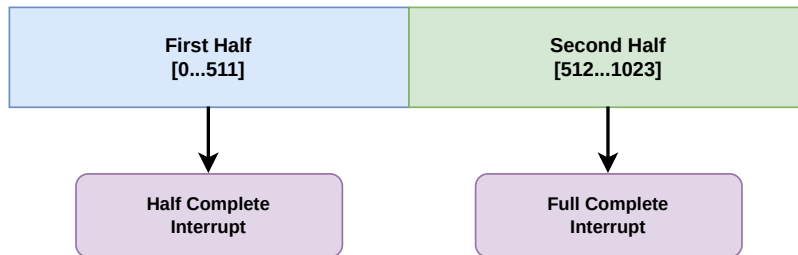
Mode	Master Transmit
Standard	Philips I2S
Data Format	16-bit
Audio Freq	48K requested → 46.875 kHz actual
MCLK Output	Enabled
DMA Mode	Circular
DMA Buffer	1024 × int16 (256 stereo frames × 2 halves)

DMA Interrupt-Driven Processing

ADC Buffer (512 × uint16):



I2S Buffer (1024 × int16 = 512 stereo L/R frames):

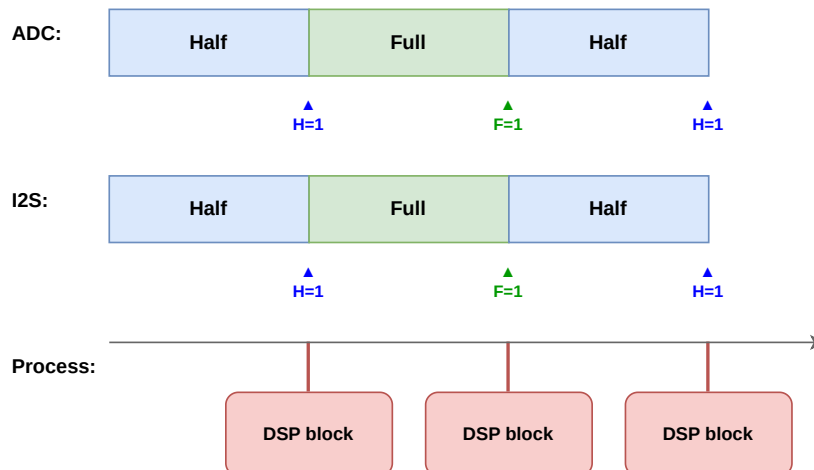


Synchronization Logic:

```
if (adc_half_ready && i2s_half_ready)
    Process: adc[0..255] → i2s[0..511]

if (adc_full_ready && i2s_full_ready)
    Process: adc[256..511] → i2s[512..1023]
```

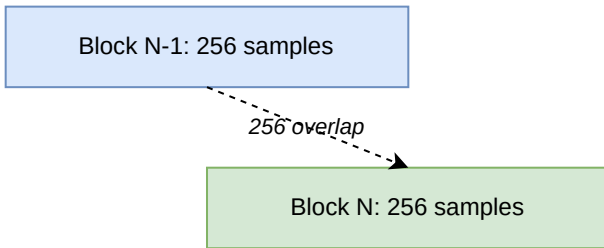
Timeline:



Lockstep: TIM2 (ADC trigger) and I2S3 both run at 46.875 kHz from the same 8 MHz crystal, so ADC and I2S flags arrive together and never drift apart.
Each half = 256 samples = **5.46 ms** — the time budget for one DSP block.
Callbacks fire when a half is *finished*: ADC half → adc[0..255] is full; I2S half → i2s[0..511] has been sent and can be refilled while DMA plays the other half.

Overlap-Add FFT Processing

Time Domain Input (512 samples total)



Why the window is applied twice

Analysis window (before FFT) limits spectral leakage.
Synthesis window (after IFFT) fades each block's edges so the mask can't cause clicks at block boundaries.

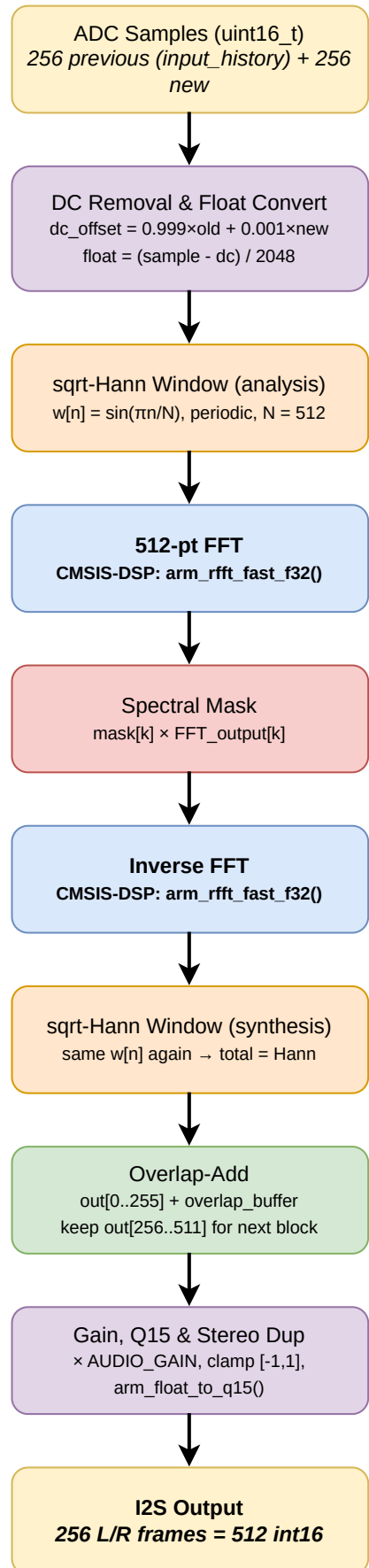
Both are $\sin(\pi n/N)$, so the total is $\sin^2(\pi n/N)$ = periodic Hann, and two Hann windows 50% apart sum to **exactly 1** (COLA): unity gain, no ripple.

Bug fixed: the first version used plain Hann both times (= Hann²). Its overlap sum swings 0.5 → 1.0 every hop, an audible 46875 / 256 ≈ **183 Hz buzz**.

Timing @ 46.875 kHz

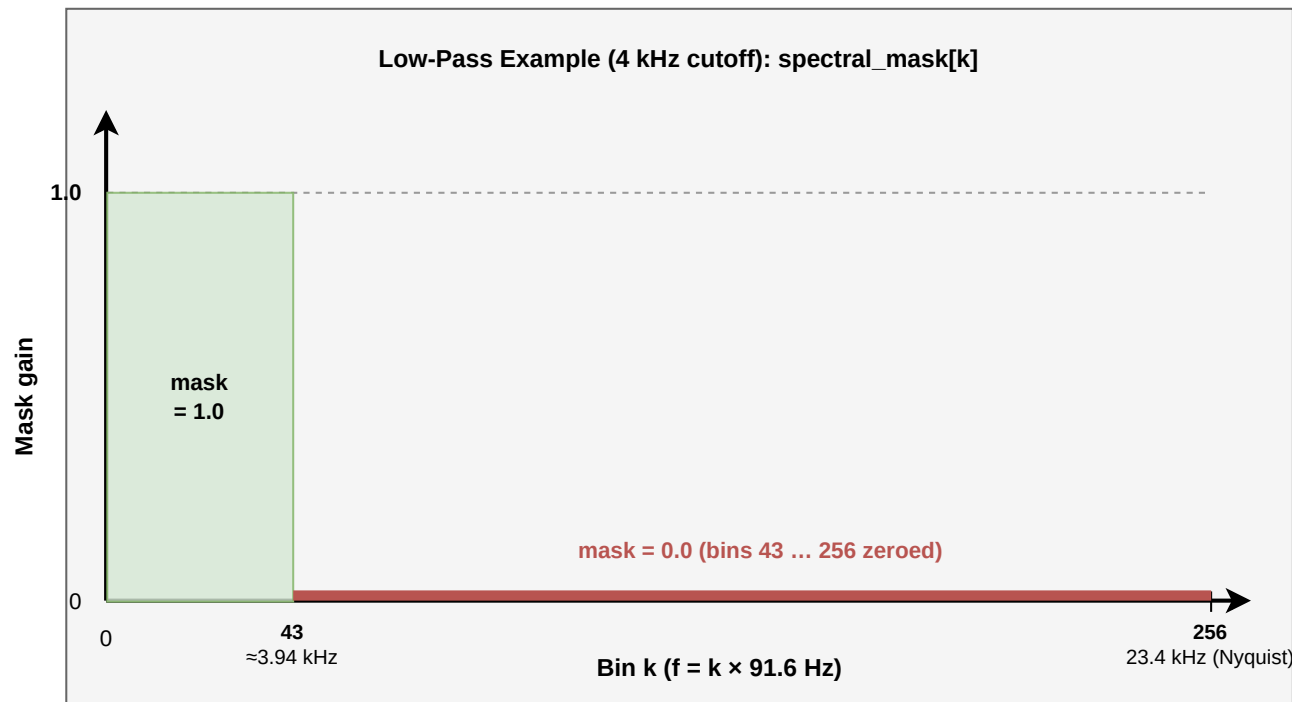
Hop 256 → new block every 5.46 ms
FFT 512 → ~10.9 ms latency
Bin width 46875 / 512 ≈ 91.6 Hz

Processing Flow:



Frequency Domain Filtering

FFT Bin Resolution: $46875 \text{ Hz} / 512 \approx 91.6 \text{ Hz/bin}$ · bins 0...256 (DC...Nyquist 23.4 kHz)



Filter Examples (bin = $(\text{int})(f \times \text{FFT_SIZE} / \text{SAMPLE_RATE})$, rounds down)	
Filter Type	Implementation
Low-Pass 4 kHz	$4000 \times 512 / 46875 = 43.7 \rightarrow \text{bin } 43$ Zero bins 43 ... 256
Band-Pass 300 – 3400 Hz	Keep bins 3 ... 37 (voice band) $300 \rightarrow 3.3 \rightarrow 3, 3400 \rightarrow 37.1 \rightarrow 37$
Notch 1 kHz whine	$1000 \times 512 / 46875 = 10.9 \rightarrow \text{bin } 10$ Zero bins 8 ... 12 (± 2)
High-Pass 100 Hz	$100 \times 512 / 46875 = 1.09 \rightarrow \text{bin } 1$, zero bins 0 ... 1 50 Hz mains hum lands in bin 0: use high-pass, not notch

Code Snippet: Low-Pass 4 kHz

```
float cutoff_freq = 4000.0f;
int cutoff_bin = (int)(cutoff_freq * FFT_SIZE / SAMPLE_RATE); // 43
for (int k = cutoff_bin; k <= FFT_SIZE / 2; k++)
    spectral_mask[k] = 0.0f;
```

Memory Buffers (All in RAM)

Buffer Name	Size	Type	Purpose
adc_buffer	512	uint16_t	main.c, ADC DMA (2 × 256)
i2s_buffer	1024	int16_t	main.c, I2S DMA (2 × 256 L/R)
fft_input	512	float32	256 previous + 256 new
fft_output	512	float32	spectrum, then IFFT result
fft_buffer	512	float32	windowed copy (RFFT overwrites input)
overlap_buffer	256	float32	tail kept for next block
input_history	256	float32	last 256 input samples
window	512	float32	sqrt-Hann $\sin(\pi n/N)$
spectral_mask	257	float32	bins 0 ... 256 gain

Total: ~14 KB static buffers + ~2.5 KB stack per block (temp_buffer, next_overlap, temp_q15)

CMSIS-DSP Functions Used (in call order)

arm_rfft_fast_init_f32(&fft_instance, 512)

Initialize 512-point real FFT (once, in Audio_Init)

arm_mult_f32(src1, src2, dst, len)

Element-wise multiply: sqrt-Hann window before FFT and after IFFT

arm_rfft_fast_f32(&fft_instance, in, out, 0)

Forward FFT (0 = forward); uses in[] as scratch

arm_rfft_fast_f32(&fft_instance, in, out, 1)

Inverse FFT (1 = inverse)

arm_add_f32(src1, src2, dst, len)

Overlap-add with previous block's tail

arm_scale_f32(src, scale, dst, len)

Apply AUDIO_GAIN

arm_float_to_q15(src, dst, len)

Convert float [-1, 1] to int16 for I2S

Optimization: Hardware FPU enabled

Compiler flags: -mcpu=cortex-m4 -mfpu=fpv4-sp-d16 -mfloat-abi=hard · -DARM_MATH_CM4